38 Case Study 1: CRISPR Field Review (2010–2024)

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38.1 Objective

Map the structure, growth, and evolution of CRISPR gene editing research from its emergence to its current state using open bibliometric data.

38.2 Setup

```
library(tidyverse)
library(openalexR)
library(igraph)
library(tidygraph)
library(ggraph)
library(quanteda)
library(tidytext)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

38.3 Data acquisition

```
works ← fetch_openalex(
  entity = "works",
  search = "CRISPR",
  from_publication_date = "2010-01-01",
  to_publication_date = "2024-06-30",
  type = "article",
  options = list(sample = 500, seed = 42)
)

works ← works ▷
  mutate(year = year(publication_date))

cat(glue("CRISPR articles retrieved: {nrow(works)}\n"))
```

```
#> CRISPR articles retrieved: 500
```

```
cat(glue("Year range: {min(works$year)}--{max(works$year)}\n"))
```

```
#> Year range: 2010--2024
```

38.4 Publication growth

```
works |>
  count(year) |>
  ggplot(aes(x = year, y = n)) +
  geom_col(fill = palette_sci(1)) +
  labs(x = "Year", y = "Publications") +
  theme_sci()
```

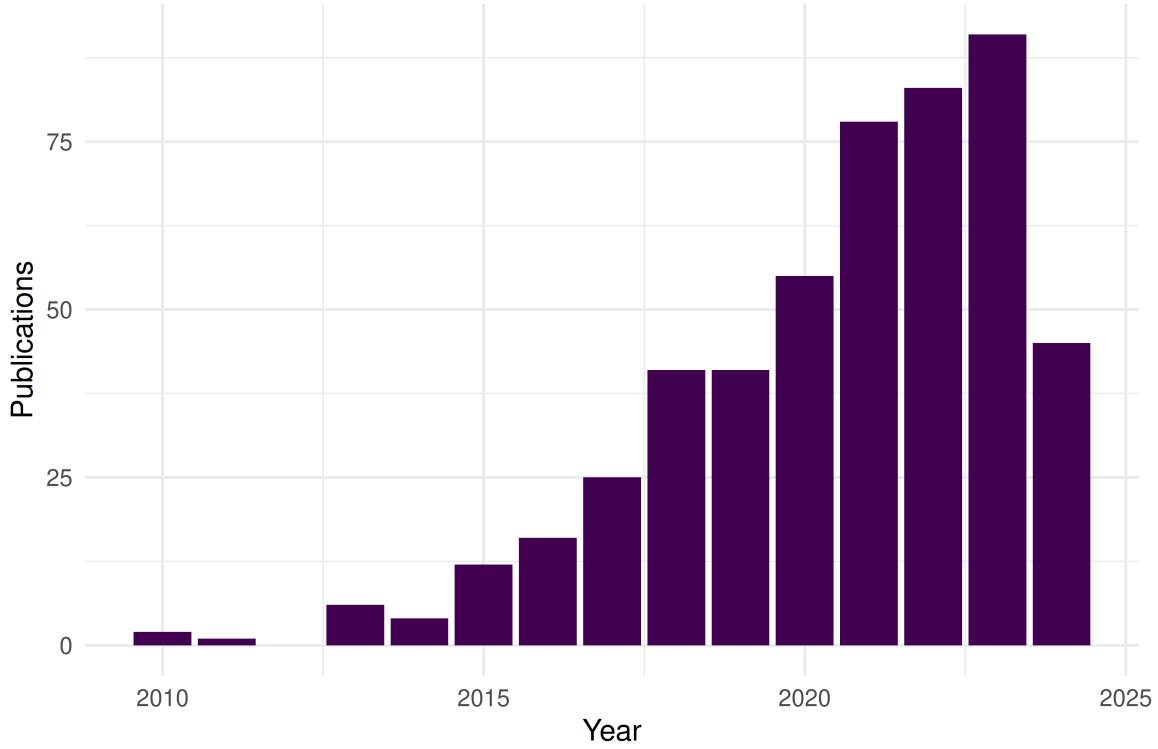


Figure 38.1: Annual publication output in the CRISPR field.

38.5 Citation landscape

```
ggplot(works, aes(x = cited_by_count)) +
  geom_histogram(binwidth = 10, fill = palette_sci(1), colour = "white") +
  labs(x = "Citations", y = "Papers") +
  theme_sci()
```

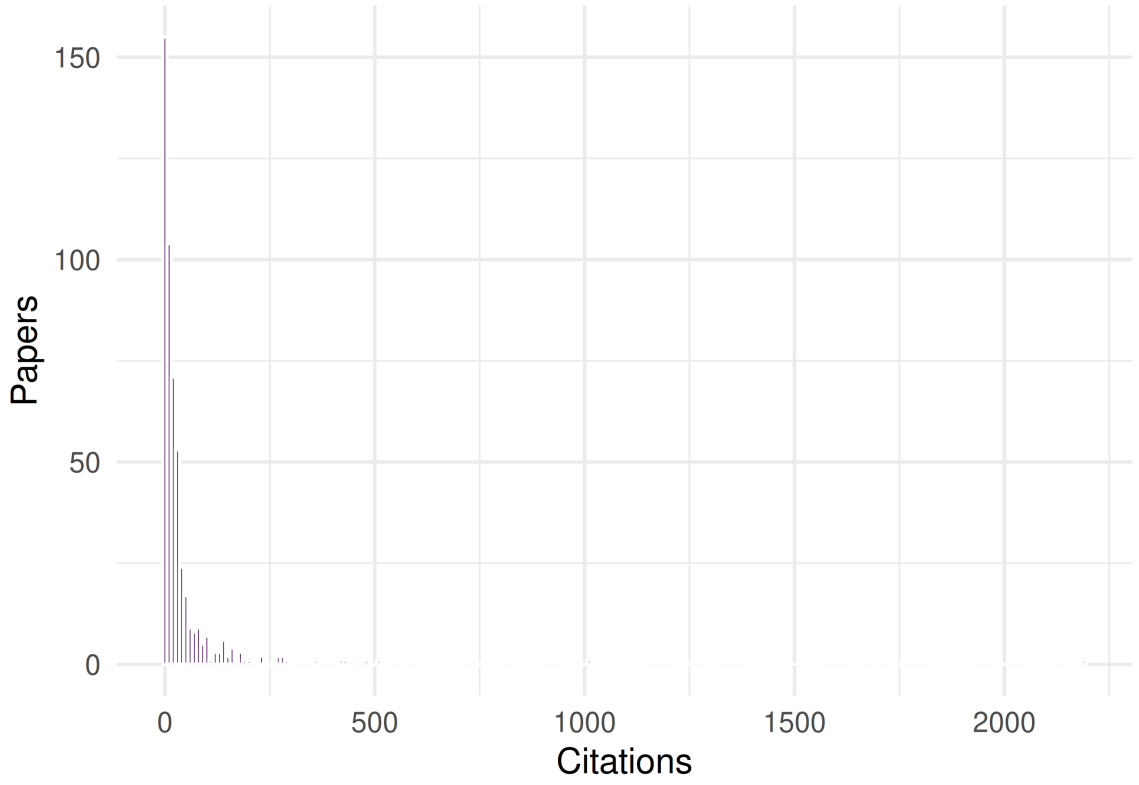


Figure 38.2: Citation distribution of CRISPR articles.

```
works ▷
  arrange(desc(cited_by_count)) ▷
  head(10) ▷
  select(display_name, year, cited_by_count, source_display_name) ▷
  gt()
```

display_name	year	cited_by_count	source_display_name
SMART: recent updates, new developments and status in 2020	2020	2195	Nucleic Acids Research
Condensin-driven remodelling of X chromosome topology during dosage compensation	2015	1013	Nature
RNA transcription modulates phase transition-driven nuclear body assembly	2015	506	Proceedings of the National Academy of Sciences

display_name	year	cited_by_count	source_display_name
Pan-cancer analysis of whole genomes identifies driver rearrangements promoted by LINE-1 retrotransposition	2020	478	Nature Genetics
Engineering of CRISPR/Cas9-mediated potyvirus resistance in transgene-free Arabidopsis plants	2016	430	Molecular Plant Pathology
Dendrimer-RNA nanoparticles generate protective immunity against lethal Ebola, H1N1 influenza, and Toxoplasma gondii challenges with a single dose	2016	421	Proceedings of the National Academy of Sciences
Characterization of Genomic Deletion Efficiency Mediated by Clustered Regularly Interspaced Palindromic Repeats (CRISPR)/Cas9 Nuclease System in Mammalian Cells*	2014	359	Journal of Biological Chemistry
CRISPR/Cas9-Mediated Genome Editing of Herpesviruses Limits Productive and Latent Infections	2016	289	PLoS Pathogens
Identification of the gene that codes for the σ 2 receptor	2017	278	Proceedings of the National Academy of Sciences
The neurotranscriptome of the Aedes aegypti mosquito	2016	276	BMC Genomics

38.6 Co-authorship network

```
author_data ← works ▷
  select(id, authorships) ▷
  unnest(authorships, names_sep = "_") ▷
  select(work_id = id, author_id = authorships_id,
         author_name = authorships_display_name) ▷
  filter(!is.na(author_id))

edges ← author_data ▷
  inner_join(author_data, by = "work_id", suffix = c("_1", "_2"),
             relationship = "many-to-many") ▷
  filter(author_id_1 < author_id_2) ▷
  count(author_id_1, author_id_2, name = "weight")

g ← graph_from_data_frame(
  edges ▷ select(author_id_1, author_id_2, weight),
  directed = FALSE
) ▷ simplify(edge.attr.comb = list(weight = "sum"))

comp ← components(g)
giant ← induced_subgraph(g, which(comp$membership == which.max(comp$csize)))
V(giant)$community ← as.factor(membership(
  cluster_leiden(giant, resolution_parameter = 1.0,
                 objective_function = "modularity")
))
V(giant)$degree ← degree(giant)

cat(glue("Network: {vcount(giant)} nodes, {ecount(giant)} edges\n"))
```

```
#> Network: 116 nodes, 3502 edges
```

```
set.seed(42)
ggraph(as_tbl_graph(giant), layout = "fr") +
  geom_edge_link(alpha = 0.1, colour = "grey60") +
  geom_node_point(aes(size = degree, colour = community), alpha = 0.7) +
  scale_size_continuous(range = c(0.5, 5), guide = "none") +
  scale_colour_manual(values = palette_sci(n_distinct(V(giant)$community))) +
  labs(colour = "Community") +
  theme_void(base_family = "sans", base_size = 11) + theme(legend.position = "bottom")
```

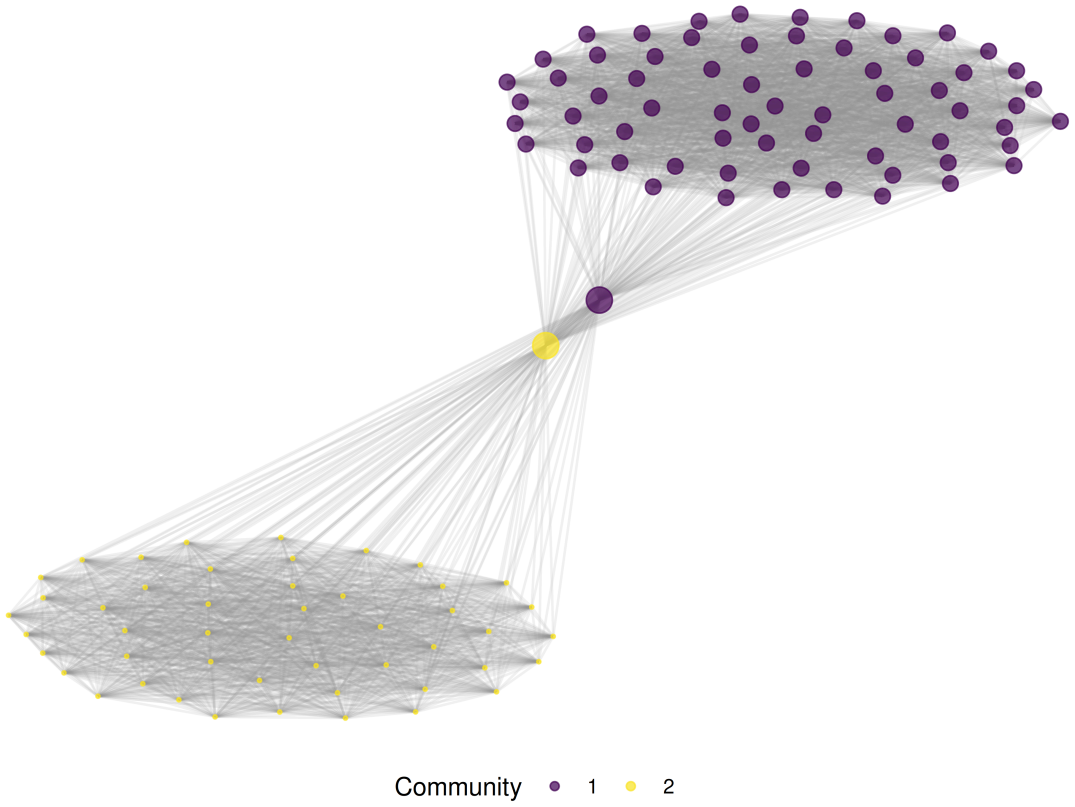


Figure 38.3: Co-authorship network of CRISPR researchers.

38.7 Topic evolution

```
text_df <- works |>
  filter(!is.na(abstract), nchar(abstract) > 50) |>
  transmute(doc_id = id, text = paste(display_name, abstract, sep = ". "), year)

corp <- corpus(text_df, docid_field = "doc_id", text_field = "text")
toks <- tokens(corp, remove_punct = TRUE, remove_numbers = TRUE) |>
  tokens_tolower() |>
  tokens_remove(stopwords("en")) |>
  tokens_remove(c("study", "paper", "results", "using", "based"))

dfmat <- dfm(toks) |> dfm_trim(min_termfreq = 5, min_docfreq = 3)

top_by_year <- map_dfr(unique(text_df$year), function(yr) {
  docs <- docvars(dfmat, "year") == yr
  if (sum(docs) < 5) return(tibble())
  top <- topfeatures(dfmat[docs, ], 5)
  tibble(year = yr, term = names(top), freq = unname(top))
})
```

```
top_by_year |>
  group_by(year) |>
  mutate(term = reorder_within(term, freq, year)) |>
  ggplot(aes(x = freq, y = term)) +
  geom_col(fill = palette_sci(1)) +
  facet_wrap(~ year, scales = "free_y", ncol = 4) +
  scale_y_reordered() +
  labs(x = "Frequency", y = NULL) +
  theme_sci(base_size = 8)
```



Figure 38.4: Top terms by year showing topical evolution.

38.8 Key findings

1. **Explosive growth**: CRISPR publications grew exponentially from 2012, reflecting the rapid adoption of Cas9-based editing.
2. **Citation concentration**: A small number of foundational papers dominate the citation landscape.
3. **Collaborative structure**: The co-authorship network shows distinct communities, likely corresponding to different application domains (therapeutics, agriculture, basic biology).
4. **Topic evolution**: Early terms focus on methodology; later years shift toward applications and clinical translation.

38.9 Lessons learned

- OpenAlex sampling provides a representative snapshot but may miss some highly specialised or non-English publications.
- The citation distribution is extreme: median citations are far below the mean, making median-based statistics essential.
- Co-authorship networks in fast-growing fields are fragmented; many research groups work independently.